
AMR-X PROTOTYPE V1 · MECHANICAL ENGINEERING

Payload-Deck Suspension

Shock Isolation for the Payload Plate: Options, Trade-offs, Recommendation and Procurement

100 kg payload · 600–800 mm deck plate · impact-dominated duty · vibration + curb-drop secondary

List of Abbreviations

Abbreviation	Meaning
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AMR	Autonomous Mobile Robot
AMR-X	Designation of this prototype platform (the robot studied in this report)
AGV	Automated Guided Vehicle
WRI	Wire Rope Isolator (looped stranded-cable shock/vibration mount)
PU	Polyurethane (elastomer / wheel material)
CoG	Centre of Gravity
RFQ	Request for Quotation
MOQ	Minimum Order Quantity
NVH	Noise, Vibration and Harshness
USPTO	United States Patent and Trademark Office
MIL-STD	United States Military Standard (e.g. MIL-STD-810 environmental)

Units

N / kN	Newton / kilonewton (force)
J	Joule (energy)
kg	Kilogram
mm	Millimetre
Hz	Hertz (natural frequency)
g	Acceleration relative to gravity ($\approx 9.81 \text{ m/s}^2$)
m/s	Metres per second

1. Why this question exists

The payload deck currently bolts **rigidly** to the lift/frame. That is fine until the moment a $\approx 100 \text{ kg}$ payload is *set down* on it. The kinetic energy of that placement has to go somewhere, and on a rigid deck it goes straight into the frame, the fasteners and the lift mechanism as a sharp force spike.

The governing relation is simple and unforgiving:

$$\text{peak force into the frame} \approx \frac{\text{impact energy}}{\text{stroke}}.$$

A feel for the numbers: 100 kg arriving with even $\approx 25 \text{ mm}$ of drop is $\approx 25 \text{ J}$. Absorbed over 25 mm of suspension travel that averages $\approx 1 \text{ kN}$ ($\approx 100 \text{ kgf}$). Absorbed over the $1\text{--}2 \text{ mm}$ of “give” a rigid bolted deck has, the same 25 J becomes **$12\text{--}25 \text{ kN}$** slammed into the structure. Interposing controlled travel buys an order-of-magnitude ($\approx 10\times$) reduction in peak force. **That force reduction — not ride comfort — is the whole point of this mechanism.**

The duty, in priority order.

- **Primary: placement shock.** Protect the robot when the payload lands on the deck.
- **Secondary: floor vibration and curb drops** transmitted up through the chassis while driving.

The catch a spring alone cannot solve. A pure spring (rubber, coil or air) *stores* the placement energy and throws it straight back — the payload bounces and you get repeated impacts. For placement shock the mechanism must **dissipate** energy, not just deflect. This single fact drives the entire ranking that follows.

Design target. Impact-dominated, with the payload varying $0 \rightarrow 100 \text{ kg}$ every cycle, on an indoor warehouse floor (small repeated unevenness — joints, ramps, dock-plate lips — not rough terrain). Crucially the deck must still present a *repeatable height* for docking and pick/place;

production AMRs in this class hold $\approx \pm 5\text{mm}$ positioning, and AGV practice warns that an unconstrained compliant deck “floats” its height and breaks dock/sensor alignment [1, 13]. The real question is therefore: **how do we put controlled, damped travel between deck and frame without losing deck-height repeatability.**

2. Candidate mechanisms

Six approaches span the spectrum from doing nothing to a fully guided, damped platform.

Option	How it works	Energy handling	Complexity	Fit
A. Rigid deck	plate bolted hard to frame	Poor (no stroke)	Lowest	Gentle hand-loading only
B. Elastomeric mounts	rubber sand-wich/cone pads	Fair; rebounds	Low	Vibration, light shock
C. Coil-spring isolators	steel springs in housings	Fair; stores & rebounds	Low	Low-freq vibration
D. Air springs / pneumatic	rubber bellows on air	Good vibration; needs air	Med	Vibration isolation
E. Wire rope isolators	looped cable, high internal (hysteretic) damping	Good; damps, low rebound	Low	Our case
F. Guided deck + hydraulic damper	plate on linear guides + shock absorber	Excellent; near-zero rebound	Med–High	Our case (worst drops)

The two that fit us are E and F. **E** (wire rope isolators) puts high energy absorption per millimetre of travel into a single bolt-on metal part: the cable strands rub and dissipate the impact instead of springing it back, and the same part also covers the secondary vibration and curb-drop duty in all three axes. **F** (a plate on linear guides with hydraulic shock absorbers) gives the cleanest

possible *no-rebound* stop for genuine drops, because a hydraulic damper converts the impact almost entirely to heat at a near-constant low force. In practice the two combine well — wire rope isolators as the primary mounts, with a rubber end-stop, or the guided-deck variant when loads are dropped rather than placed.

Options B, C and D are kept as the *vibration* layer, not the impact layer: a single-rate rubber, coil or air spring sized to catch a 100 kg drop is far too stiff when the deck is empty, and rubber can take a set or tear under repeated heavy impacts.

3. Real-product precedents

Option E — wire rope isolators. These are the defence- and aerospace-standard answer to *repeated* shock precisely because they damp in all three axes, are unaffected by oil, temperature and ageing, and are maintenance-free; the standard catalogue lines are qualified to environmental standards such as MIL-STD-810 [2]. The recurring design rule is to mount them at $\approx 45^\circ$ to the load, which equalises the stiffness across all three axes so one part handles the vertical placement shock *and* the horizontal curb/stop loads [3, 4]. The category is already used in automated-warehouse equipment [2].

Option F — guided deck with hydraulic shock absorbers. Industrial self-compensating shock absorbers convert impact to heat and do not rebound. The decisive feature for us is that the payload energy changes every cycle as the load varies $0 \rightarrow 100$ kg, and self-compensating units “react to changing energy conditions without adjustment”, covering effective weights from ≈ 1 kg to 1,550 kg with a side-load adapter for impact angles up to 25° [5, 6]. Mounted under a linearly-guided plate with light return springs, they form the no-bounce drop-catcher of Solution F.

Why not a bare spring. Coil, rubber and air mounts are excellent *vibration* isolators — air springs in particular reach very low natural frequencies [10, 12] — but used alone for placement shock they store the energy and bounce the load. And on an AGV/AMR specifically, soft unconstrained compliance under the deck varies the deck height and “breaks sensor and dock-plate alignment”, which is why such compliance is normally reserved for manual carts unless its travel is constrained [13].

4. Recommendation: Solution E (wire rope isolators)

We recommend Solution E — mounting the deck plate on wire rope isolators at the four corners. It attacks our exact failure mode (placement energy spiking into the frame) at the source, it damps rather than rebounds, and it delivers the primary *and* both secondary duties in one maintenance-free, bolt-on metal part with no air supply or power. It is also the only one of the two fitting options available as a *finished, purchasable component*, which de-risks and accelerates the prototype.

4.1 The trade-off you must accept

A compliant deck trades structural protection against deck-height repeatability. The same travel that protects the frame also lets the deck move under load, so dock/pick height can drift and the payload can sway in turns. Accept this by **constraining the travel**: hard limit stops plus a rubber end-stop, modest stroke (~ 15 – 25 mm), and isolators placed symmetrically about the payload CoG so each corner carries an equal ≈ 25 kg share. If a *precise* docking height is required, decouple the two needs in time — soft-isolate while driving, then mechanically lock or clamp the deck during pick/place and docking.

If loads can be genuinely *dropped* (not gently lowered), step up to Solution F: a guided deck on linear bearings with hydraulic self-compensating absorbers and light return springs. It gives the cleanest no-rebound stop, but — like a fabricated rocker in the drivetrain study — it is an *assembly*, not a single purchasable kit.

5. Specifications to preserve or improve

Any deck-suspension solution must **meet or beat** the rigid baseline on the points below. Targets derived from the impact and packaging study:

Parameter	Current (rigid deck)	Requirement
Static payload	100 kg	support 100 kg + dynamic margin
Per-corner static load	≈ 25 kg (4 mounts)	isolator at $\sim 50\%$ of rating (shock headroom)
Vertical travel	≈ 0 (rigid)	~ 15 – 25 mm <i>controlled</i> , with hard stops
Peak force into frame	12–25 kN on a drop	cut $\approx 10\times$ by adding stroke
Rebound after placement	n/a	damped, no bounce (dissipative)
Load variability	n/a	cope $0 \rightarrow 100$ kg per cycle
Deck-height repeatability	fixed by bolt-up	maintain for docking (limit stops / lock)
Environment	indoor warehouse	oil/temp/age-stable, maintenance-free

A 4-corner wire rope set clears these with margin; the items needing active confirmation in the RFQ are the **drop height / placement velocity** (which fixes the energy and therefore the isolator size and count) and the **travel-limit / end-stop** scheme that protects deck-height repeatability.

6. Ready-to-buy products suited to our use case

The following are appropriate for a 100 kg payload deck (with the noted configuration). All are catalogue parts with low MOQ.

Product	Why it fits us / what to configure	Type
Enidine WR6-700-10 [7]	≈ 50.8 kg/unit, ≈ 43 mm tall. Four corners at 45° ; at 25 kg/corner it sits at $\sim 50\%$ rating, leaving compression travel for the impact. Primary pick.	Wire rope isolator
Enidine WR8-400-08 [8]	≈ 68.9 kg/unit, ≈ 54 mm tall. Step up to this (or run six WR6) if placement drops are harsh.	Wire rope isolator
ACE SC ² self-compensating absorbers [5]	Effective weight 1–1,550 kg, side-load adapter to 25° , self-compensates as load varies $0 \rightarrow 100$ kg. The energy dump for Solution F.	Hydraulic damper
Parker LORD Flex-Bolt sandwich (small group) [9]	≈ 3 – 66 kg/mount; cheap, simple vibration layer to add beneath/alongside the primary mounts.	Elastomeric mount
ACE PAL-3 [11]	Auto-levelling air spring, ≤ 385 kg/spring; adds constant deck level + isolation if levelling is needed (requires air).	Pneumatic levelling
PGFUN WRI [14]	Low-cost 10-loop stainless isolators to validate geometry before committing to engineered units. <i>Prototype only.</i>	Wire rope (proto)

Primary picks for us: $4\times$ **Enidine WR6-700-10** (or six, or step to WR8 for harsh drops), mounted at 45° with a rubber end-stop, because they damp the placement impact, cover vibration and curb drops in the same part, need no air or power, and are maintenance-free. The single open point is the **exact size/count**, which the supplier sizing tool fixes once the drop height is known. Hold **Solution F** (guided deck + **ACE SC²**) as the upgrade for true drop loading.

7. Priority of this modification

Priority: MEDIUM now, rising to HIGH before any payload-handling trials.

- **Not a bring-up blocker.** An empty or gently hand-loaded deck on a rigid mount is fine for powering up the robot and validating electronics, navigation and the safety chain. It does not gate first motion.
- **Becomes critical for payload operations.** Once the robot does automated or repeated payload placement — and once it drives a real floor with joints, dock-plates and curb lips — the rigid deck funnels every placement and curb impact straight into the frame, fasteners and lift mechanism, where repeated multi-kN spikes drive fatigue and shock the electronics. This must be resolved before field trials.
- **Ranking against other open items.** It sits *below* the chassis-geometry and drivetrain decisions that determine whether the platform can be built at all, but it *interacts with the lift mechanism* the deck mounts to — so the suspension interface must be fixed together with the lift, not bolted on afterwards.

Recommended action: issue an RFQ / order for 4–8× Enidine WR6-700-10 (confirm exact part and count via the supplier sizing tool once drop height/velocity is measured), mount them at 45° with hard limit stops and a rubber end-stop, and keep the guided-deck + ACE SC² assembly (Solution F) as the fallback for genuine drop loading.

References

- [1] ABB. *AMR P603V Product Specification* (3HKA00000226401) — lift-type AMR; deck raised on ball-screw lifters; positioning to $\approx \pm 5$ mm; on-deck load-measurement sensors.
- [2] Enidine (ITT). *Wire Rope Isolators* — stainless cable, RoHS aluminium bars; qualified to MIL-STD-810, BV43-44, MIL-STD-167 and similar; used in automated-warehouse equipment. <https://www.enidine.com/en-us/products/wire-rope-isolators>
- [3] USPTO. *Multi-axis shock and vibration isolation system*, US 6,530,563 (wire rope isolators at $\approx 45^\circ$ to the suspended mass for balanced three-axis stiffness). <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/6530563>
- [4] USPTO. *Shock resistant device*, US 7,260,476 (mounting wire rope isolators at $\approx 45^\circ$ equalises compression/shear/roll stiffness and extends life). <https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/7260476>
- [5] ACE Controls. *SC² Self-Compensating Industrial Shock Absorbers* — effective weight 1–1,550 kg, integrated positive stop, side-load adapter to 25°, react to changing energy without adjustment. <https://www.acecontrols.com/us/>
- [6] ACE Controls / Valin. *Heavy / self-compensating industrial shock absorbers* (self-compensating units react to changing energy conditions without adjustment). <https://www.valin.com/motion-control-and-automation/products/heavy-industrial-shock-absorbers-ace-controls-inc>
- [7] Enidine. *WR6-700-10 wire rope isolator datasheet* (load capacity ≈ 50.8 kg / 112 lb; thickness ≈ 43 mm). <https://datasheets.globalspec.com/ds/itt-inc/wr6-700-10/36ff8e41-4c53-43df-81f6-df1d272cb7a7>
- [8] Enidine. *WR8-400-08 wire rope isolator datasheet* (load capacity ≈ 68.9 kg / 152 lb; thickness ≈ 54 mm). <https://datasheets.globalspec.com/ds/itt-inc/wr8-400-08/07d5da4c-76cb-48fa-be8d-d096f6d5a05b>
- [9] Parker LORD. *Flex-Bolt Sandwich Mounts* (small group ≈ 31 – 645 N ≈ 3 – 66 kg per mount; metric thread). <https://www.modusadvanced.com/parker-lord-vibration-isolation-mounts>
- [10] Firestone Industrial. *Airstroke / Airmount Engineering Manual & Design Guide* (air-spring isolators; highest isolation of any type; no spring chatter). <https://www.firestoneairide.com/content/dam/fsip/pdfs/airstroke/Actuators-and-Isolators-Metric-Design-Guide.pdf>

- [11] ACE Controls. *PAL-3 to PAL-9 low-frequency pneumatic levelling mounts* (automatic level control; up to 385 kg per air spring; natural frequency to ~ 1.7 Hz). <https://www.acecontrols.com/us/products/vibration-control/low-freq-pneumatic-levelling-mounts.html>
- [12] Mason Industries. *Spring Isolators / housed spring mounts* (steel-coil low-frequency vibration mounts; horizontal stiffness drops with deflection — add limit stops). <https://mason-ind.com/spring-isolators/>
- [13] CasterHQ. *Spring-loaded casters — AGV guidance* (sprung compliance under a deck introduces variable deck height that breaks sensor / dock-plate alignment unless travel is constrained). <https://casterhq.com/collections/casters-spring-loaded>
- [14] PGFUN. *10-loop 304 stainless wire rope isolator* (low-cost prototype isolators; more loops give better cushioning). <https://www.amazon.com/PGFUN-Isolator-Stainless-Anti-Vibration-Absorber/dp/B09W695DJ6>